AWS Cloud Practitioner — Exam Cheat Sheet

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1. Trigger Word Mapping

> Grouped by category. Scan the category header first, then find your keyword.

🖥️ Compute

If you see this keyword...	...the AWS service is this
"Serverless functions / event-driven / runs infrequently / short bursts"	AWS Lambda
"Serverless containers / no EC2 to manage"	AWS Fargate
"Docker / **containers** / managed orchestration"	Amazon **ECS**
"Kubernetes workload / managed Kubernetes"	Amazon EKS
"Auto launch/terminate EC2 to match demand"	Amazon EC2 Auto Scaling
"Distribute incoming traffic across instances / AZs"	Elastic Load Balancing (ELB)
"Large-scale parallel batch compute jobs / simulations running hours"	AWS Batch
"Simplest website / WordPress / beginner / minimal cloud experience"	Amazon Lightsail
"Platform as a Service / upload code / AWS manages platform"	AWS Elastic Beanstalk
"Deploy containerized web app from source / minimal overhead"	AWS App Runner
"Prepackaged EC2 template / OS + software / launch template"	Amazon Machine Image (AMI)
"Physical isolation / BYOL per-socket or per-core licensing"	Dedicated Hosts
"Fault-tolerant / stateless / can be interrupted / up to 90% off"	Spot Instances
"Steady-state / 1-year commitment / no interruptions"	Reserved Instances
"Partial Upfront + 1 year + no interruption / best RI discount"	Partial Upfront Reserved Instances
"EC2 + Lambda long-term flexible savings"	Compute Savings Plans

If you see this keyword...	...the AWS service is this
"Billing per second / Amazon Linux exact time"	On-Demand (per-second billing)
"Layer 7 / URL path-based routing / HTTP & HTTPS/Least effort"	Application Load Balancer

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🗄️ Databases

If you see this keyword...	...the AWS service is this
"Relational DB / transactional / order system / SQL / managed"	Amazon RDS
"Oracle / SQL Server / MySQL / PostgreSQL / lift-and-shift DB"	Amazon RDS
"AWS-native relational / MySQL or PostgreSQL compatible only"	Amazon Aurora
"Database HA / automatic failover / standby in another AZ"	Amazon RDS Multi-AZ
"Scale database reads / analytics offload / read-only copy"	Amazon RDS Read Replica
"NoSQL / key-value / millions of req/sec / single-digit ms"	Amazon DynamoDB
"Active-active NoSQL across multiple AWS Regions"	Amazon DynamoDB Global Tables
"In-memory DynamoDB cache / microsecond latency"	Amazon DynamoDB Accelerator (DAX)
"OLAP / analytics / data warehouse / complex SQL on petabytes"	Amazon Redshift
"Serverless Redshift / no cluster to manage"	Amazon Redshift Serverless
"In-memory / caching / sub-millisecond reads / Redis or Memcached"	Amazon ElastiCache
"Graph database / relationship-heavy data / social network"	Amazon Neptune
"MongoDB / JSON document storage"	Amazon DocumentDB
"Apache Cassandra / wide-column NoSQL"	Amazon Keyspaces
"Automatic storage cost savings / unknown or changing access patterns"	S3 Intelligent-Tiering
"Time-series database / IoT telemetry / trillions of events per day"	Amazon Timestream
"Migrate a database to AWS with source still operational"	AWS Database Migration Service (DMS)
"View specific data / Keep control over full datasets / Manage access for many customers"	S3 Access Points

- HA -> 2 + AZs
- Max. Resiliency-> 3 + AZs

🪣 Storage

If you see this keyword...	...the AWS service is this
"Object storage / durable / offsite backup / static website / data lake"	Amazon S3
"Long-term archive / lowest cost storage / rarely accessed"	S3 Glacier Deep Archive
"Infrequent access / multi-AZ / immediately retrievable"	S3 Standard-IA
"Infrequent access / single AZ / cheapest IA / recreatable data"	S3 One Zone-IA
"WORM / prevent S3 object deletion / immutable objects"	Amazon S3 Object Lock
"Versioning / recover deleted S3 objects / previous versions"	Amazon S3 Versioning
"Replicate S3 to another region / cross-region DR"	S3 Cross-Region Replication (CRR)
"Fast long-distance S3 uploads / speed up transfers via edge"	Amazon S3 Transfer Acceleration
"Block storage / attached to one EC2 / persistent hard drive"	Amazon EBS
"Temporary local block storage / lost when instance stops"	EC2 Instance Store
"Shared file system / NFS / hundreds of EC2 instances concurrently"	Amazon EFS
"Infrequent access / multi-AZ EFS / lower cost"	EFS Standard-IA
"Windows file system / SMB / Active Directory integration"	Amazon FSx for Windows
"High-performance compute file system / HPC / Lustre"	Amazon FSx for Lustre
"Tape library replacement / existing backup workflow unchanged"	AWS Storage Gateway — Tape Gateway
"NFS or SMB access to S3 / file gateway hybrid on-premises"	AWS Storage Gateway — File Gateway
"Block storage backed by S3 / **iSCSI** / hybrid"	AWS Storage Gateway — **Volume** Gateway
"Physical offline data migration / ~80 TB per device"	AWS Snowball Edge
"Exabyte / 100+ petabytes offline migration / truck-sized"	AWS Snowmobile
"Centralized backup policy across multiple services (RDS, EFS, EBS...)"	AWS Backup
"Automate EBS snapshot lifecycle / EBS only"	Amazon Data Lifecycle Manager (DLM)

🔒 Security & Identity

If you see this keyword...	...the AWS service is this
"SQL injection / XSS / custom web rules / filter HTTP traffic"	AWS WAF
"DDoS protection / L3/L4 / always on / free"	AWS Shield Standard
"DDoS + response team + cost protection / advanced DDoS"	AWS Shield Advanced
"Centrally manage WAF + Shield across accounts"	AWS Firewall Manager
"Threat detection / network + account monitoring / ML-based"	Amazon GuardDuty
"Sensitive data / PII discovery in S3"	Amazon Macie
"Vulnerability scanning / EC2 software flaws / CVEs"	Amazon Inspector
"Root cause investigation of security incidents"	Amazon Detective
"Aggregate security findings / single pane of glass"	AWS Security Hub
"Download SOC / ISO / PCI compliance reports / audit agreements"	AWS Artifact
"Encryption keys/ Cryptographic / multi-tenant / integrates with all AWS services"	AWS KMS
"Single-tenant HSM / FIPS 140-2 Level 3 / strict compliance"	AWS CloudHSM
"Store and auto-rotate DB passwords / secrets"	AWS Secrets Manager
"Store config values / app config / no auto-rotation"	AWS Systems Manager Parameter Store
"Free TLS/SSL certificates for CloudFront / ALB / AWS services"	AWS Certificate Manager (ACM)
"S3 encryption at rest by AWS / server-side"	SSE-S3 or SSE-KMS
"Social login / Facebook / Google / SAML for customer-facing apps"	Amazon Cognito
"SSO for AWS accounts + SaaS apps / employee portal"	AWS IAM Identity Center
"Managed Microsoft Active Directory in the cloud"	AWS Directory Service
"Cross-account access / temporary permissions for EC2 / services"	IAM Role
"Temporary federated credentials / short-term tokens"	AWS STS
"Programmatic access / CLI / SDK / API authentication"	Access Keys
"Find overly permissive policies / unused IAM permissions"	IAM Access Analyzer
"Audit all IAM users / key age / MFA status in one report"	IAM Credential Report
"Assign permissions to multiple IAM users at once"	IAM Groups

If you see this keyword...	...the AWS service is this
"Best practice checks / unrestricted SSH / service limits/Qoutas/cost Provisioning"	AWS Trusted Advisor
"Resource config history / what changed on a resource"	AWS Config
"Who did what / API call history / console login audit"	AWS CloudTrail
"VPC traffic capture(RDS,Aurora,Elasticache,ELB) / network interface **IP** metadata"	VPC Flow Logs
"Illegal use of AWS / report abuse"	AWS Abuse Team

- Who enable Encryption Keys? -> Customers
- What ? -> KMS
- TLS/SSL -> encrypt data in transient
- KMS -> ..... at rest

🌐 Networking

If you see this keyword...	...the AWS service is this
"Multiple isolated networks in same account"	Amazon VPC
"**Subnet**-level firewall / stateless"	Network ACLs
"Instance-level firewall / stateful"	Security Groups
"Make subnet public / route to internet"	Internet Gateway + Route Tables
"Private subnet internet access / outbound only"	NAT Gateway
"Dedicated private connection to AWS / no internet / high speed"	AWS Direct Connect
"Encrypted tunnel over public internet / quick setup"	AWS Site-to-Site VPN
"DNS / domain routing / highly available scalable DNS"	Amazon Route 53
"Static files globally / CDN / cache at **edge** / low latency worldwide"	Amazon CloudFront
"Accelerate global traffic / non-HTTP / route via AWS private backbone"	AWS Global Accelerator
"Connect multiple VPCs + on-premises / hub-and-spoke"	AWS Transit Gateway
"Direct 1:1 private connection between two VPCs (same Region)"	VPC Peering
"Access S3/DynamoDB from private subnet without internet"	VPC Endpoint (Gateway) — Free
"Private IP access to AWS services from VPC / PrivateLink"	VPC Endpoint (Interface)

If you see this keyword...	...the AWS service is this
"5G edge latency / embedded in carrier network"	AWS Wavelength
"Low-latency in a specific major city / metro area"	AWS Local Zones
"Physical AWS hardware in YOUR data center / data residency / true hybrid"	AWS Outposts
"Store data in specific country / legal data residency"	AWS Regions
"Share AWS resources (Aurora+VPC) across Acc. "	AWS Resource Access Manager (RAM)

💰 Cost Management

If you see this keyword...	...the AWS service is this
"Estimate costs BEFORE deployment / migration cost model"	AWS Pricing Calculator
"On-premises assessment / migration business case / cost projection"	AWS Migration Evaluator
"Visualize / manage EXISTING spend over time / forecasts"	AWS Cost Explorer
"Set alerts / budget thresholds / proactive spend notifications"	AWS Budgets
"Most granular billing data / raw CSV export to S3"	AWS Cost and Usage Report (CUR)
"Consolidated billing / multiple accounts / share RIs"	AWS Organizations
"Bill departments within one account / attribute costs by team / AWS Generated"	Cost Allocation Tags
"Rightsize EC2 / ML-based recommendations / historical usage"	AWS Compute Optimizer
"View and request increases for service limits"	AWS Service Quotas
"Warn when approaching service limits / best practice checks / Reduce costs / Identifying **underutilized** resources"	AWS Trusted Advisor
"Carbon emissions / environmental impact report"	AWS Customer Carbon Footprint Tool
"View of Estimates on metered to date"	Billing Dashboard

📊 Analytics & AI/ML

If you see this keyword...	...the AWS service is this
"Serverless SQL on S3 / CSV files / query without infrastructure"	Amazon Athena
"Business intelligence / dashboards / charts / visualizations"	Amazon QuickSight
"ETL / extract transform load / data catalog / serverless"	AWS Glue
"Big data cluster / Hadoop / Spark / managed EMR"	Amazon EMR
"Real-time **data streaming** / ingest high-volume events / replay"	Amazon **Kinesis** Data Streams
"Build and train custom ML models / requires ML expertise"	Amazon SageMaker
"Enterprise ML-powered document / content search / no ML expertise"	Amazon Kendra
"Image / video analysis / content moderation / no ML expertise"	Amazon Rekognition
"Extract text from documents / invoices / scanned forms"	Amazon Textract
"Speech to text / transcribe call recording"	Amazon Transcribe
"Text to speech / lifelike audio from text"	Amazon Polly
"Chatbot / conversational interface / voice + text / Alexa engine"	Amazon Lex
"Text sentiment / NLP / entity detection / language analysis"	Amazon Comprehend
"Product recommendations / personalization / real-time"	Amazon Personalize
"Forecasting / time-series business predictions"	Amazon Forecast
"Human review for low-confidence ML predictions"	Amazon Augmented AI (A2I)

📨 Messaging & Integration

If you see this keyword...	...the AWS service is this
"Pub/sub / SMS + email + Lambda simultaneously / fan-out"	Amazon SNS
"Decouple / queue / async messaging / one consumer per message"	Amazon SQS
"Email delivery only / transactional / bulk email / no SMS"	Amazon Simple Email Service (SES)
"Schedule a Lambda / CRON job / event bus / no custom code"	Amazon EventBridge
"Orchestrate Lambda functions / multi-step visual workflow"	AWS Step Functions
"Real-time data streaming / replay / multiple consumers"	Amazon Kinesis Data Streams

If you see this keyword...	...the AWS service is this
"Legacy message broker / ActiveMQ or RabbitMQ migration"	Amazon MQ
"Managed GraphQL API / connect multiple backends"	AWS AppSync
"Fully managed virtual desktop / complete cloud computer for employees"	Amazon WorkSpaces/AppStream

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🛠️ Developer & Infrastructure Tools

If you see this keyword...	...the AWS service is this
"Infrastructure as Code / YAML templates / repeatable stacks"	AWS CloudFormation
"CloudFormation across multiple accounts and regions"	AWS CloudFormation StackSets
"Define IaC with Python / Java / TypeScript"	AWS Cloud Development Kit (CDK)
"Cloud-based IDE / write run debug in browser"	AWS Cloud9
"Browser-based shell / CLI without local install / zero setup"	AWS CloudShell
"CI/CD pipeline / full build-test-deploy orchestration"	AWS CodePipeline
"Compile and test code / managed build service"	AWS CodeBuild
"Automate code deployment to EC2 / Lambda / on-premises"	AWS CodeDeploy
"Private Git repository on AWS"	AWS CodeCommit
"Centralize operational data / Automate tasks across EC2"	AWS Systems Manager
"Automate OS **patching EC2** + on-premises servers"	AWS Systems Manager Patch Manager
"Browser-based SSH / no SSH keys / run commands remotely"	AWS Systems Manager Session Manager
"Centralized parameter storage / store config values"	AWS Systems Manager Parameter Store

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🏢 Governance, Migration & Support

If you see this keyword...	...the AWS service is this
"Consolidated billing / central policy / **SCPs** across accounts/ aggregate usages/RIs discount"	AWS Organizations
"Multi-account landing zone / guardrails / governance setup"	AWS Control Tower

If you see this keyword...	...the AWS service is this
"Lift-and-shift server migration to EC2 / physical VM to cloud"	AWS Application Migration Service (MGN)
"Track migration progress centrally / multiple migration tools"	AWS Migration Hub
"Identify capability gaps / cloud readiness evaluation"	AWS Cloud Adoption Framework (CAF)
"Evaluate a specific workload against AWS best practices"	AWS Well-Architected Tool
"Concierge + TAM + 24/7 + architecture guidance"	AWS Enterprise Support
"TAM only / designated technical account manager"	AWS Enterprise Support
"Health API / full Trusted Advisor checks / 24/7 phone support"	AWS Business Support
"Concierge answers billing inquiries / NOT technical support"	AWS Enterprise Support Concierge
"Global AWS service status / outage RSS feed"	AWS Health Dashboard — Service Health
"Personalized health alerts for YOUR specific resources"	AWS Health Dashboard — Your Account Health
"Buy / sell third-party software / AMIs / SaaS on AWS"	AWS Marketplace
"Paid consulting / cloud adoption outcomes / AWS own team"	AWS Professional Services
"Third-party implementation / build migrate manage workloads"	APN Consulting Partners
"Software vendor / ISV / technology product partner"	APN Technology Partners

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2. Common Distractors

> Ordered by exam frequency. Top ones appear most often.

CloudTrail vs. Config vs. CloudWatch

- CloudTrail = WHO did WHAT — API calls, console logins, who deleted a resource, when.
- Config = WHAT CHANGED — resource configuration history, compliance rules, detect drift.
- CloudWatch = PERFORMANCE — metrics, alarms, CPU spikes, logs, auto-scaling triggers.
- Trap: "who terminated the EC2?" → CloudTrail. "When was the EBS volume removed?" → Config. "CPU over 80% alert" → CloudWatch.

WAF vs. Shield Standard vs. Shield Advanced

- WAF = L7 application layer; blocks SQL injection, XSS, custom IP/geo rules; you write the rules.

- Shield Standard = L3/L4 network layer; automatic DDoS protection; free for all customers; always on.
- Shield Advanced = paid; adds 24/7 DDoS response team (DRT) + cost protection + app-layer monitoring.
- Trap: "SQL injection" → WAF. "DDoS" → Shield. "SQL injection AND DDoS" → WAF + Shield Advanced.

Security Groups vs. Network ACLs

- Security Groups = EC2 instance level; stateful (return traffic automatic); allow rules only (by default all blocked).
- Network ACLs = subnet level; stateless (must allow both directions); allow AND deny rules.
- Trap: "VPC subnet firewall" → ACL. "EC2 instance firewall / specific instance rules" → Security Group.

Spot vs. On-Demand vs. Reserved vs. Savings Plans vs. Dedicated

- Spot = up to 90% off; CAN be interrupted; fault-tolerant / stateless / batch workloads only.
- On-Demand = no commitment; highest hourly rate; for unpredictable or short-term workloads.
- Reserved = 1–3 year commitment; up to 72% off; for steady, predictable production workloads.
- Savings Plans = flexible commitment on EC2 + Lambda + Fargate; more flexible than Reserved.
- Dedicated Instances = dedicated hardware per account; does NOT support BYOL per-socket/core licensing.
- Dedicated Hosts = full physical server; supports BYOL per-socket/per-core; most expensive.
- Trap: "fault-tolerant batch" → Spot. "1-year stable production" → Reserved. "BYOL Windows Server" → Dedicated Hosts (not Dedicated Instances).
- DISASTER RECOVERY FOR EC2 (AMIs & EBS snapshots)

RDS Multi-AZ vs. RDS Read Replica

- Multi-AZ = synchronous Replication in another AZ; automatic failover; HIGH AVAILABILITY only; does NOT improve read performance.
- Read Replica = asynchronous read-only copy; READ SCALABILITY and reporting; NOT an HA failover target.
- Trap: "disaster recovery / failover" → Multi-AZ. "Scale reads / offload analytics" → Read Replica.

Pricing Calculator vs. Cost Explorer vs. Budgets vs. Migration Evaluator

- Pricing Calculator = estimate costs BEFORE you build; model hypothetical workloads.
- Migration Evaluator = analyze on-premises usage to produce a migration cost estimate.
- Cost Explorer = visualize and forecast costs you are ALREADY spending; historical + 12-month forecast.
- Budgets = set thresholds; receive alerts when actual or forecasted spend crosses the limit.

- Trap: "before migration / pre-deployment estimate" → Pricing Calculator or Migration Evaluator. "Already on AWS / visualize spend" → Cost Explorer. "Alert me when I exceed $X" → Budgets.

GuardDuty vs. Macie vs. Inspector vs. Security Hub vs. Detective

- GuardDuty = THREATS — ML detects malicious behavior in CloudTrail, VPC Flow Logs, DNS.
- Macie = PII DATA — discovers and classifies sensitive data in S3.
- Inspector = VULNERABILITIES — scans EC2, Lambda, and containers for CVEs and software flaws.
- Security Hub = AGGREGATOR — single dashboard for findings from GuardDuty, Inspector, Macie, etc.
- Detective = ROOT CAUSE — investigates HOW a threat happened using graph analytics.
- Trap: "continuously scan EC2 for software vulnerabilities" → Inspector. "Detect malicious account activity" → GuardDuty. "Find PII in S3" → Macie. "All security findings in one place" → Security Hub.

Cognito vs. IAM Identity Center vs. IAM

- Cognito = CUSTOMERS — external app users; social login (Facebook, Google, Amazon).
- IAM Identity Center = EMPLOYEES — workforce SSO for AWS accounts and enterprise SaaS apps.
- IAM = INTERNAL — manages users, roles, and permissions within an AWS account.
- Trap: "social media login for app users" → Cognito. "Employee AWS Console SSO" → IAM Identity Center. "Grant EC2 access to S3" → IAM Role.

Outposts vs. Local Zones vs. Wavelength

- Outposts = physical AWS rack INSIDE YOUR data center; for data residency, legal obligations, on-site processing.
- Local Zones = AWS-managed infrastructure in a major city; extends a Region; for latency-sensitive urban apps.
- Wavelength = embedded INSIDE 5G carrier networks; for ultra-low-latency mobile apps.
- Trap: "slow internet / must stay on-premises / data residency" → Outposts. "Near large city" → Local Zones. "5G mobile edge" → Wavelength.

Snowball vs. Snowmobile

- Snowball Edge = physical device ~80 TB per unit; for typical offline data migration; Remote + Processing (Ship in sea)
- Snowmobile = ruggedized shipping container truck; for 100+ petabytes / exabyte-scale migrations.
- Trap: "50 petabytes" → Snowmobile. "10 TB–few PB" → Snowball Edge.

SNS vs. SQS vs. SES vs. EventBridge vs. Kinesis

- SNS = push / pub-sub / fan-out ONE message to MANY protocols simultaneously (email, SMS, Lambda, SQS).
- SQS = pull / queue / ONE consumer per message; for decoupling microservices.

- SES = email ONLY; for transactional, marketing, and bulk email delivery to end users.
- EventBridge = event BUS; routes events between services; runs scheduled CRON jobs; no custom code.
- Kinesis = real-time high-throughput DATA STREAMING; multiple consumers; data is replayable.
- Trap: "SMS + email fan-out" → SNS. "Decouple services async" → SQS. "Email only" → SES. "Schedule Lambda daily" → EventBridge. "Real-time streaming" → Kinesis.

Step Functions vs. CodePipeline vs. EventBridge

- Step Functions = orchestrates Lambda functions into a sequenced multi-step serverless workflow.
- CodePipeline = CI/CD pipeline for code releases: Source → Build → Test → Deploy.
- EventBridge = event bus or scheduler; routes events or runs CRON jobs; no orchestration logic.

CloudFront vs. Global Accelerator

- CloudFront = CDN; caches CONTENT (HTML, images, video) at edge; for static/dynamic content delivery.
- Global Accelerator = network-layer; NO caching; routes TCP/UDP traffic to nearest healthy Region over AWS private backbone.
- Trap: "cache static files globally" → CloudFront. "Improve TCP app performance globally / non-HTTP" → Global Accelerator.

Secrets Manager vs. Parameter Store vs. KMS

- Secrets Manager = stores + auto-rotates credentials (DB passwords, API keys); integrates with RDS; has a cost.
- Parameter Store = stores configuration values; no native auto-rotation; free tier available.
- KMS = manages ENCRYPTION KEYS; not a secrets store; used BY Secrets Manager to encrypt secrets.

ECS vs. EKS vs. Fargate vs. ECR vs. App Runner

- ECR = container image REGISTRY; stores Docker images only.
- ECS = container ORCHESTRATION; AWS-native; simpler than Kubernetes.
- EKS = managed Kubernetes; for teams already using Kubernetes.
- Fargate = serverless COMPUTE ENGINE for ECS or EKS; no EC2 to manage.
- App Runner = simplest; source code or container image → running HTTPS web app automatically.
- Trap: "store Docker images" → ECR. "Run containers, no servers" → Fargate. "Kubernetes" → EKS. "Simplest container deploy" → App Runner.

DynamoDB Global Tables vs. DynamoDB DAX

- Global Tables = active-active multi-region replication; low-latency reads AND writes from any region.

- DAX = in-memory microsecond cache in front of DynamoDB; single-region; read-only acceleration.
- Trap: "write from multiple regions" → Global Tables. "Faster reads same region" → DAX.

Control Tower vs. Organizations

- Control Tower = automated landing zone for NEW multi-account environments; uses SCPs + Config rules as guardrails.
- Organizations = consolidated billing + SCP management for EXISTING accounts; no automated setup.

Transit Gateway vs. VPC Peering

- Transit Gateway = hub-and-spoke; connects MANY VPCs + on-premises; supports transitive routing.
- VPC Peering = direct 1:1 private connection between exactly two VPCs; non-transitive.

SageMaker vs. Pre-trained AI Services

- SageMaker = build + train CUSTOM ML models; requires ML expertise.
- Pre-trained (Rekognition, Textract, Transcribe, Polly, Lex, Comprehend, Personalize, Forecast, Kendra) = no ML expertise needed.
- Trap: "no ML expertise" → always pick a pre-trained service, never SageMaker.

Direct Connect vs. Site-to-Site VPN

- Direct Connect = dedicated PRIVATE physical line; no internet; consistent bandwidth; for compliance or high-throughput.
- Site-to-Site VPN = encrypted tunnel OVER the public internet; quick setup; variable latency; lower cost.
- Trap: "must NOT go over internet" → Direct Connect. "Quick encrypted connection" → VPN.

Multi-AZ vs. Multi-Region

- Multi-AZ = within ONE Region; protects against AZ failure; HIGH AVAILABILITY.
- Multi-Region = across 2+ Regions; protects against full Region failure; DISASTER RECOVERY + global users.

VPC Endpoint (Gateway) vs. NAT Gateway

- VPC Endpoint Gateway = FREE; connects private subnet to S3 or DynamoDB without internet.
- NAT Gateway = allows private subnet instances to reach the internet outbound; NOT for private AWS service access.

Health Dashboard: Service Health vs. Your Account Health

- Service Health = global status of ALL AWS services; RSS feed; widespread outages.

- Your Account Health = personalized alerts for events impacting YOUR specific running resources.

AWS Batch vs. Lambda

- Batch = LONG-RUNNING parallel batch jobs (minutes to hours); manages compute fleet automatically.
- Lambda = SHORT event-driven functions (max 15 minutes); serverless; triggered by events.
- Trap: "simulations running up to 3 hours" → Batch. "Process S3 upload event" → Lambda.

DLM vs. AWS Backup

- DLM = automates EBS snapshot lifecycle ONLY.
- AWS Backup = centralized policy-driven backup for MULTIPLE services (RDS, EFS, DynamoDB, S3, EBS, EC2).

CodeBuild vs. CodeDeploy vs. CodePipeline

- CodeBuild = BUILD stage; compiles code and runs tests.
- CodeDeploy = DEPLOY stage; deploys code to EC2, Lambda, on-premises.
- CodePipeline = ORCHESTRATES the full pipeline: Source → Build → Test → Deploy.

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3. Shared Responsibility Matrix

Responsibility Area	AWS ✅	Customer ✅
Physical hardware & data centers	✅	
Network infrastructure & virtualization layer	✅	
Hypervisor patching	✅	
Decommissioning end-of-life storage devices	✅	
Availability Zone & Region availability	✅	
Managed service OS patching (RDS, Aurora, Lambda, DynamoDB)	✅	
Aurora / RDS database engine patching	✅	
Lambda OS, runtime, and underlying infrastructure	✅	
Patch management — host OS & hypervisor	✅	
Configuration management — AWS infrastructure layer	✅	
Security awareness & training (AWS staff)	✅	
EC2 guest OS patching & updates		✅

Responsibility Area	AWS ✅	Customer ✅
Application code security		✅
IAM users, roles, policies, and MFA enablement		✅
Data encryption — configuring and enabling it		✅
Managing the encryption process for clusters and snapshots	✅	
EBS volume backups		✅
Security group & network ACL configuration		✅
Database encryption settings (enabling on Aurora/RDS)		✅
S3 bucket ACLs, bucket policies, and permissions		✅
Lambda: writing the code + security inside the code		✅
DynamoDB access permissions (IAM policies)		✅
Patch management — guest OS (EC2) & customer applications		✅
Configuration management — customer apps & data		✅
Choice of AWS Region where data is stored		✅
Security awareness & training (customer staff)		✅

Shared Controls — both AWS AND customer manage their own layer

- Patch Management = AWS patches the host; customer patches the guest OS and apps.
- Configuration Management = AWS manages the infrastructure layer; customer manages their app and data configs.
- Awareness & Training = AWS trains its staff; customer trains their own staff.

Root User Tasks — Customer only, never delegate

- Change or cancel the AWS Support plan
- Close the AWS account
- Create an AWS Organization
- Enable MFA on the root account
- Restore IAM user permissions after accidental lockout
- No need for explicit permission
- Trap: everything else should be done with IAM users or roles, NEVER root for daily tasks.
- Trap: AWS Organization Admin can limit priviliges of root user

Shared Responsibility by Service Type

- IaaS (EC2) = Customer manages most: OS patching, middleware, app, data, firewall config.

- PaaS (RDS, Elastic Beanstalk) = AWS manages OS and runtime; customer manages data, app, and access.
- SaaS (DynamoDB, S3, Lambda) = AWS manages everything infrastructure; customer manages data, access, and code logic only.

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4. Service Comparisons

Compute Decision Tree

- Full OS control / lift-and-shift → EC2
- Short event-driven function / max 15 min / no servers → Lambda
- Containers without managing servers → Fargate
- Containers with OS-level access → ECS on EC2
- Kubernetes workload → EKS
- Simplest website / beginner / WordPress → Lightsail
- Upload code, AWS manages platform → Elastic Beanstalk
- Source code → HTTPS app, fully managed → App Runner
- Long batch jobs / parallel simulations → AWS Batch
- like App store for users -> service catalog

Database Decision Tree

- SQL / relational / transactional → RDS
- Fastest relational / Most Scalable DB / MySQL or PostgreSQL only → Aurora
- NoSQL / key-value / massive scale → DynamoDB
- Analytics / OLAP / complex SQL on large datasets / Columnar Storage / Auto. backups → Redshift
- In-memory caching / sub-millisecond → ElastiCache
- Graph / relationships → Neptune
- MongoDB / JSON documents → DocumentDB
- Cassandra / wide-column → Keyspaces
- Time-series / IoT / high-volume events → Timestream
- RDS High Availability / failover → Multi-AZ
- RDS Read Scaling / analytics offload → Read Replica
- DynamoDB global reads and writes → Global Tables
- DynamoDB faster reads same region → DAX

Storage Decision Tree

- Unstructured objects / backups / data lake → S3
- Persistent block storage / single EC2 → EBS
- Temporary local storage / high I/O → EC2 Instance Store
- Shared file system / multiple EC2 → EFS

- Windows file system / Active Directory → FSx for Windows
- HPC / Lustre → FSx for Lustre
- Archive / long-term / rarely accessed → S3 Glacier Deep Archive
- Infrequent access / multi-AZ → S3 Standard-IA
- SCT convert DB schemas & Code Objects

Support Plans

- Basic = free; documentation, forums, 7 core Trusted Advisor checks; NO human support.
- Developer = adds email support during business hours; 1 primary contact.
- Business = adds 24/7 phone/chat/email; ALL Trusted Advisor checks; AWS Health API; 1 hour prod. down
- Enterprise On-Ramp = adds pool of TAMs; Concierge; 30-min critical response.
- Enterprise = adds designated TAM; Concierge; 15-min critical response; architecture reviews.
- Concierge team = answers BILLING and ACCOUNT inquiries ONLY; NOT technical support.

Well-Architected Framework — 6 Pillars

Pillar	One-Line Focus	Key Design Principles
Operational Excellence	Run and improve workloads	Make small reversible changes; perform ops as code
Security	Protect data and systems	Enable traceability; least privilege; encrypt everything
Reliability	Recover from failure	Function correctly & consistently;Auto-recover; test recovery; multiple AZs; stop guessing capacity
Performance Efficiency	Use resources efficiently	Quickly Adv.;Go global in minutes; use serverless; use advanced tech as a service
Cost Optimization	Deliver value at lowest price	Measure efficiency; rightsize; use Spot for batch
Sustainability	Minimize environmental impact	Maximize utilization; use managed services; eliminate idle resources

AWS CAF — 6 Perspectives

Perspective	Primary Stakeholders	Core Focus
Business	CFOs, CEOs, Business Analysts	Align cloud investments with business goals
People	(Tech + People) HR, Staffing, Change Managers	Org change management; culture and skills
Governance	CIOs, Portfolio Managers	Risk management; align cloud with business strategy

Perspective	Primary Stakeholders	Core Focus
Platform	CTOs, IT Architects, Engineers	Design and deliver cloud infrastructure
Security	CISOs, Security Analysts	Data protection, compliance, incident response
Operations	IT Operations, SREs	Day-to-day cloud service delivery and monitoring

- CAF Phases: Envision → **Align** (identify capability gaps) → Launch → Scale

Global Infrastructure

- Region = geographic cluster of 2+ AZs; data does not leave a Region unless you explicitly move it.
- Availability Zone (AZ) = one or more discrete data centers in a single location; own power and networking; interconnected with other AZs via private low-latency links; do NOT share power with other AZs.
- Edge Locations = Points of Presence (PoPs) used by CloudFront and Global Accelerator; far more numerous than Regions.

S3 Quick Facts

- Total capacity = unlimited; maximum single object size = 5 TB.
- Durability = 99.999999999% (11 nines) across all classes except One Zone-IA.
- S3 Standard 99.99%
- S3 One Zone-IA = only class that does NOT replicate across multiple AZs.
- Lose 1 Obj. every 10K years if store 10M obj.

EC2 Billing Facts

- Amazon Linux / Windows On-Demand = billed per second; 60-second minimum.
- Convertible Reserved Instances = can exchange for other Convertible RIs of a different family/size/OS; CANNOT be sold on Marketplace.
- Standard Reserved Instances = CAN be sold on the Reserved Instance Marketplace; cannot change instance family.
- Dedicated Instances = dedicated hardware per account; does NOT support BYOL per-socket/core.
- Dedicated Hosts = full physical server; supports BYOL per-socket, per-core, or per-VM licensing.

Key Cloud Benefits

Exam Phrase	What It Means
Trade fixed for variable expense	Pay-as-you-go; OpEx instead of CapEx
Economies of scale	AWS buying power lowers prices for everyone
Stop guessing capacity	Elasticity; scale up/down on demand

Exam Phrase	What It Means
Increase speed and agility	Provision in minutes, not weeks
Go global in minutes	Deploy to any Region instantly
Stop spending on data centers	AWS manages physical infrastructure
Elasticity	Scale to exact demand; eliminate idle capacity
High Availability	Multiple AZs; automatic failover; no single point of failure
Fault Tolerance	System continues despite component failure
Durability	Long-term data protection (S3 = 11 nines)
Reliability	System performs its function and recovers from failure

IAM Security Best Practices

- Enable MFA for root account and ALL IAM users.
- Never use root account for daily tasks; create individual IAM users.
- Apply least privilege — grant only the permissions needed.
- Use IAM Roles for EC2 instances and cross-account access, not Access Keys.
- Rotate Access Keys regularly; delete unused keys.
- Use IAM Groups to assign permissions to multiple users at once.
- Use IAM Credential Report to audit key age and MFA status.
- Use IAM Access Analyzer to find overly permissive or unused policies.
- Store secrets in Secrets Manager — never hard-code credentials in code.

Common Exam Traps — Quick Reference

Trap Pattern	Correct Answer
"No infrastructure to manage"	Serverless: Lambda, Fargate, Athena, DynamoDB
"No ML expertise required"	Pre-trained: Rekognition, Textract, Transcribe, Polly, Lex, Comprehend, Kendra
"Least operational overhead"	Fully managed / serverless over IaaS
"Fault-tolerant batch / stateless / can be interrupted"	Spot Instances
"Consistent, stable, 1+ year production"	Reserved Instances
"Temporary / unpredictable / dev environment"	On-Demand Instances
"Physical server + BYOL per-socket licensing"	Dedicated Hosts (NOT Dedicated Instances)
"Migrate 100+ PB data offline"	Snowmobile (NOT Snowball)

Trap Pattern	Correct Answer
"Access S3 from EC2, no public internet"	VPC Endpoint Gateway (free)
"Database HA / automatic failover"	RDS Multi-AZ (NOT Read Replica)
"Scale database reads / analytics offload"	RDS Read Replica (NOT Multi-AZ)
"CloudTrail vs CloudWatch"	CloudTrail = WHO (audit). CloudWatch = WHAT (performance)
"Inspector vs GuardDuty"	Inspector = software vulnerabilities. GuardDuty = active threats
"Pricing Calculator vs Cost Explorer"	Calculator = FUTURE estimate. Explorer = PAST analysis
"Download compliance documents"	AWS Artifact
"IAM Role vs IAM User"	Role = temporary/cross-account/services. User = permanent/human
"Concierge team"	Billing and account inquiries ONLY — not technical support
"Control Tower vs Organizations"	Control Tower = new automated landing zone. Organizations = existing accounts
"Aurora vs RDS"	Aurora = MySQL or PostgreSQL ONLY. RDS = also Oracle, SQL Server, MariaDB
"Multi-AZ vs Multi-Region"	Multi-AZ = high availability. Multi-Region = disaster recovery
"Convertible vs Standard RI"	Convertible = exchange family. Standard = sell on Marketplace
"EC2 OS patching"	Always customer responsibility on EC2